# Choosing the right QA strategy for a successful project

Tuesday, April 7



# Agenda

- Why is QA the most important factor behind successful software projects?
- How to develop quality software and not to go bankrupt?
- Which types of tests will be the best for your software project?
- What are the practical differences between Quality Assurance and Quality Control?
- What is the "extended standard approach" in software testing?

How can you overcome the fear of starting a project?

The fear of:

- exceeding budget
- exceeding time
- low usability
- not achieving business goals

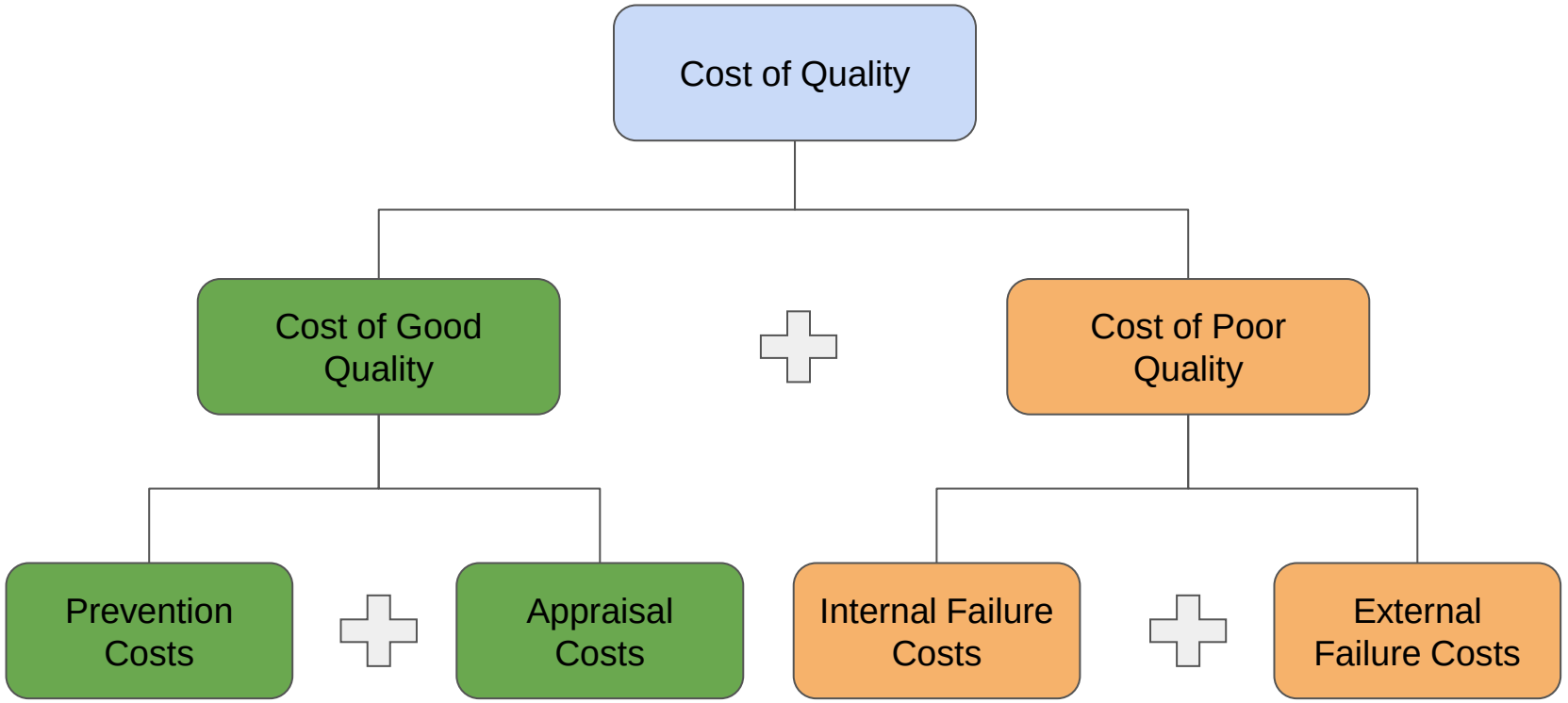
*"Avoid temptation to take a shortcut in software development process."*

# Quality - definition?

- different definition for each company
- multiple definitions in the internet

*"Quality is a **well-defined process** for creating a **useful product** that **adds value** for both the consumer and the manufacturer." - IBM*

# Cost of quality



# Cost of poor quality

Expensive software failures:

- Heathrow Terminal 5 opening - $50 milion
- Knight Capital Group - $440 milion in 30 minutes
- Patriot Missile Error - 28 lives

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# How to develop quality software and not to go bankrupt?

"Choose the right QA strategy"

- optimal preventive methods
- optimal approach to testing
- test automation approach

# Quality Assurance - the way of understanding

Misunderstanding and communication

- what I thought that QA is
- habits in decision making
- always ask questions
- you share the same goal

# QA decisions

Road through the real project

- QA role
- difference between QA and QC
- choosing the right QA strategy
- how we improved our process in one of TSH's project
- types of tests that you might have not heard about
- good and bad practices

# Do a good planning and analysis

*“By failing to prepare, you are preparing to fail.” - Benjamin Franklin*

# QA vs QC

| Quality Assurance | Quality Control |
| --- | --- |
| Proactive | Reactive |
| Process oriented | Product oriented |
| Improving development and test process | Identifying defects in product |
| Everyone responsible | Usually testers |

# QA strategy

- QA should be a member of a team since the planning stage
- QA should be responsible for following the company standards
- do not test everything
- always think about ROI - “Return of Investment”
- play the role of the end user

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# How to start the project as QA?

# Goals of automated tests

- Quick feedback
- Reliability
- Coverage

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# Choose the right tools for tests automation

- think about needs of the end user
- be open for new solutions
- do not experiment too much

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# What practices should be avoided?

- tests of external libraries
- communication is always the key

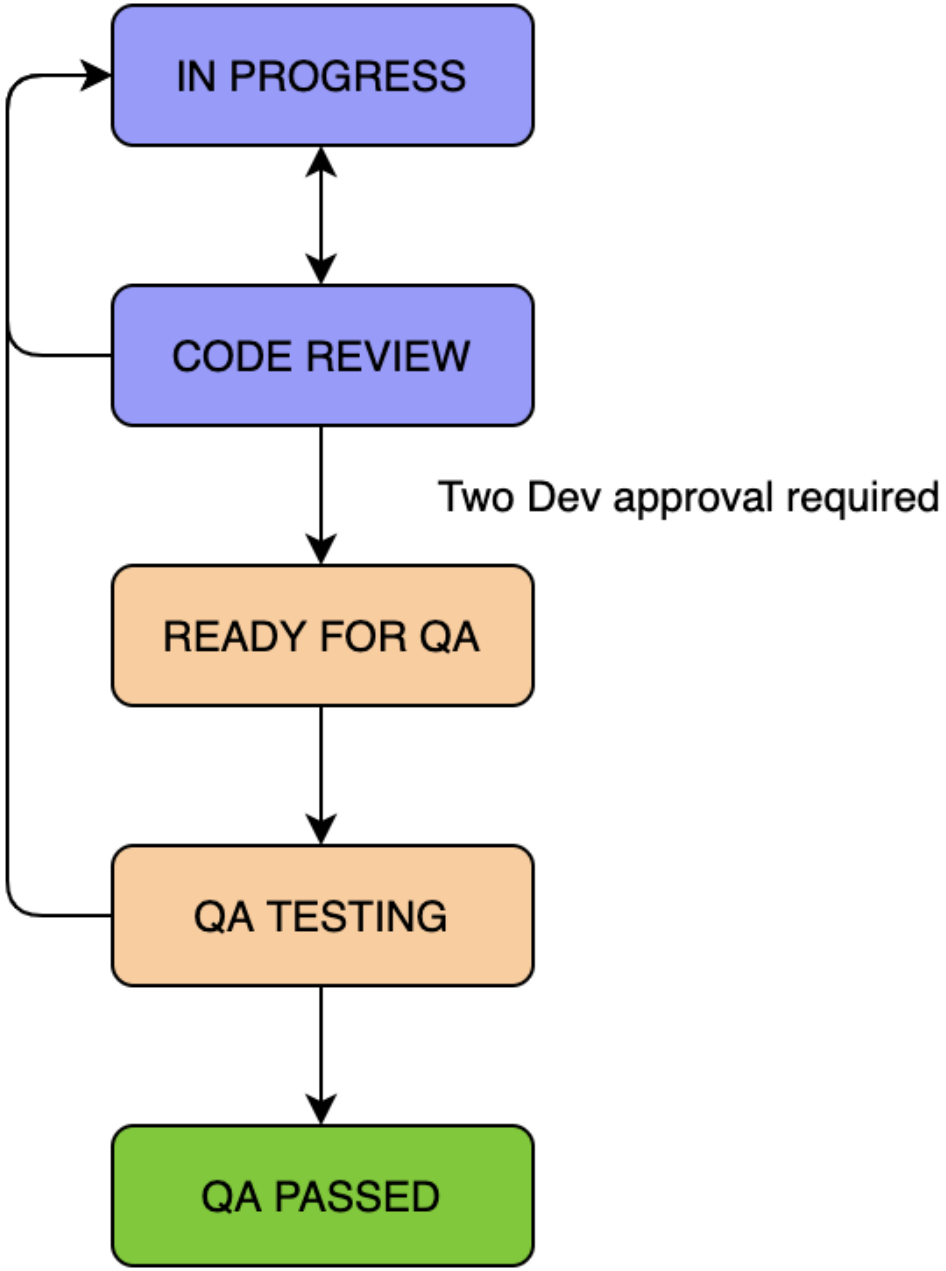
# Validate schema

What do we know about Joi?

- schema description language
- data validator for JavaScript
- the library is tested

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# Project flow



# Project example

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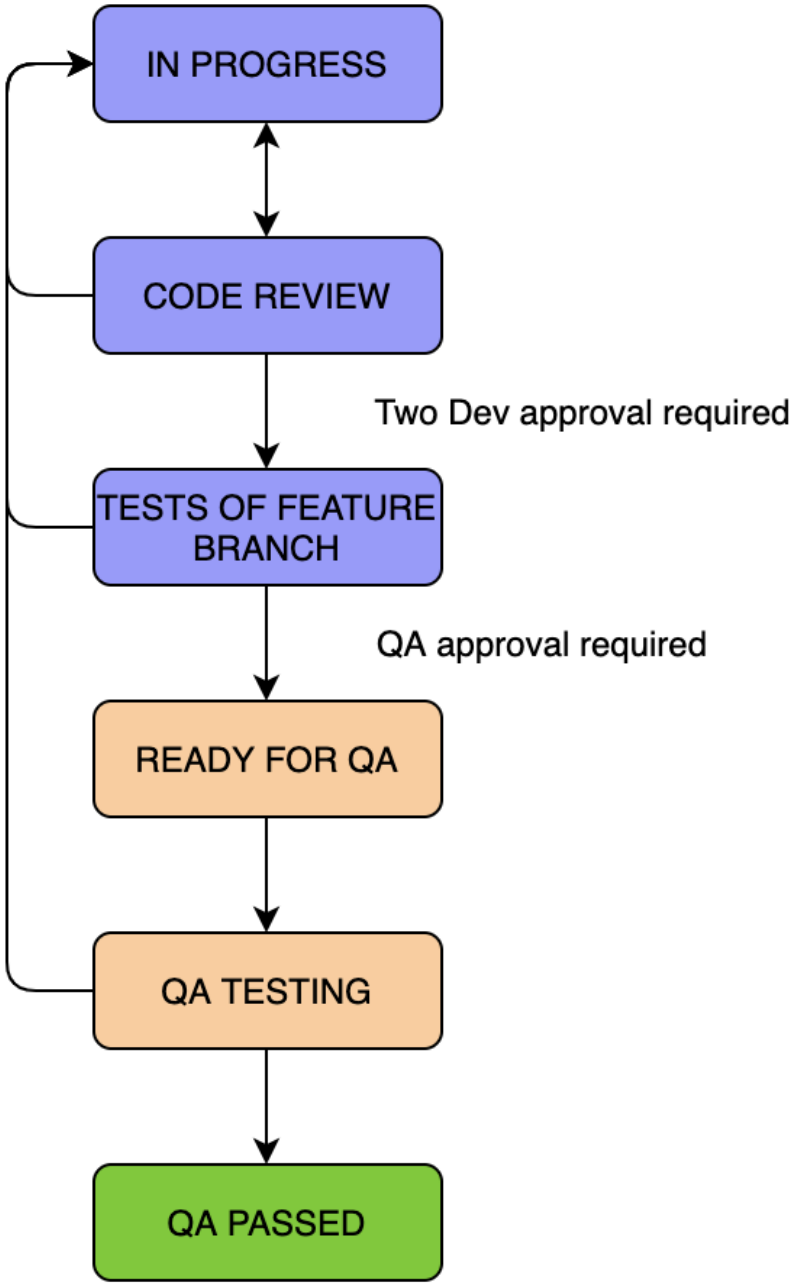
How we started?

- not synced code
- never ending problems with deployments
- bugs on the master during the sprint
- stressful development at the end of a sprint

# Project example

*“Stay committed to your decisions, but stay flexible in your approach.”*
*- Tony Robbins*

# Project flow



# Project example after the process improvements

What we changed?

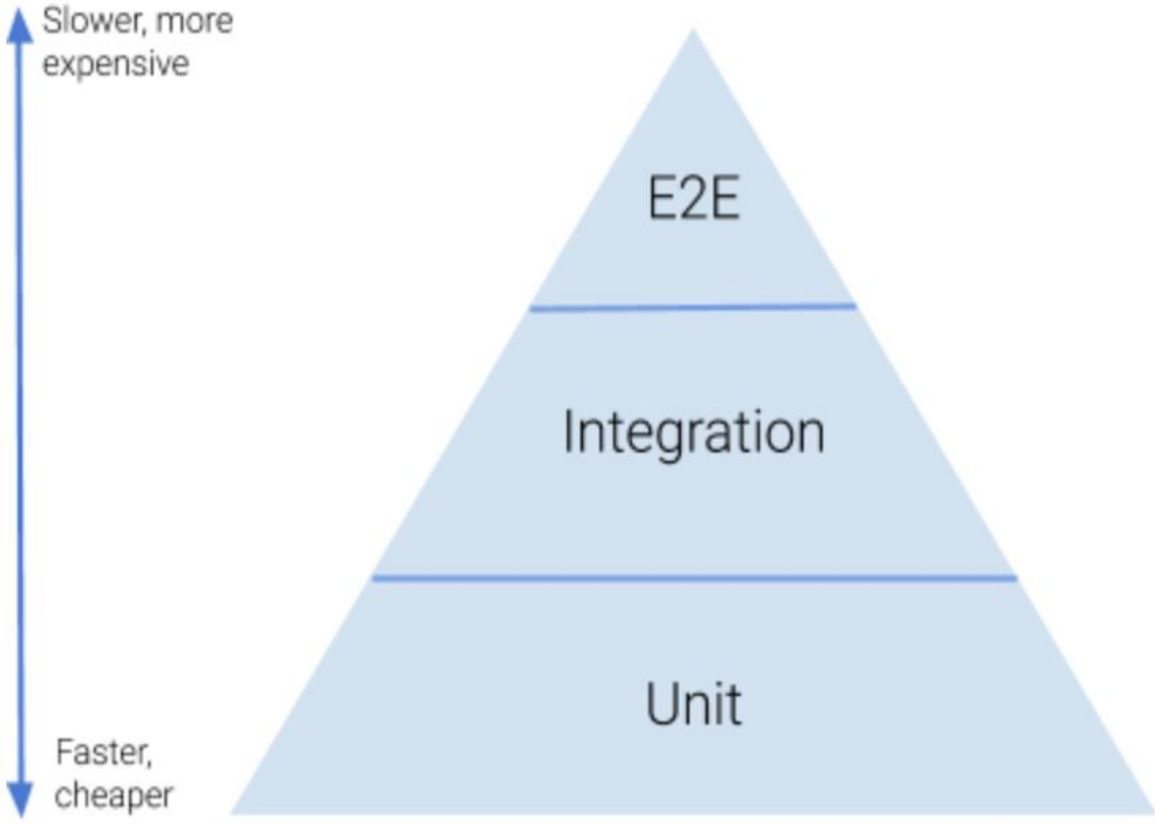
- tests on feature branches
- often deployments
- QA responsible for merging pull requests
- half-way demo
- retrospectives meeting

# What strategy?

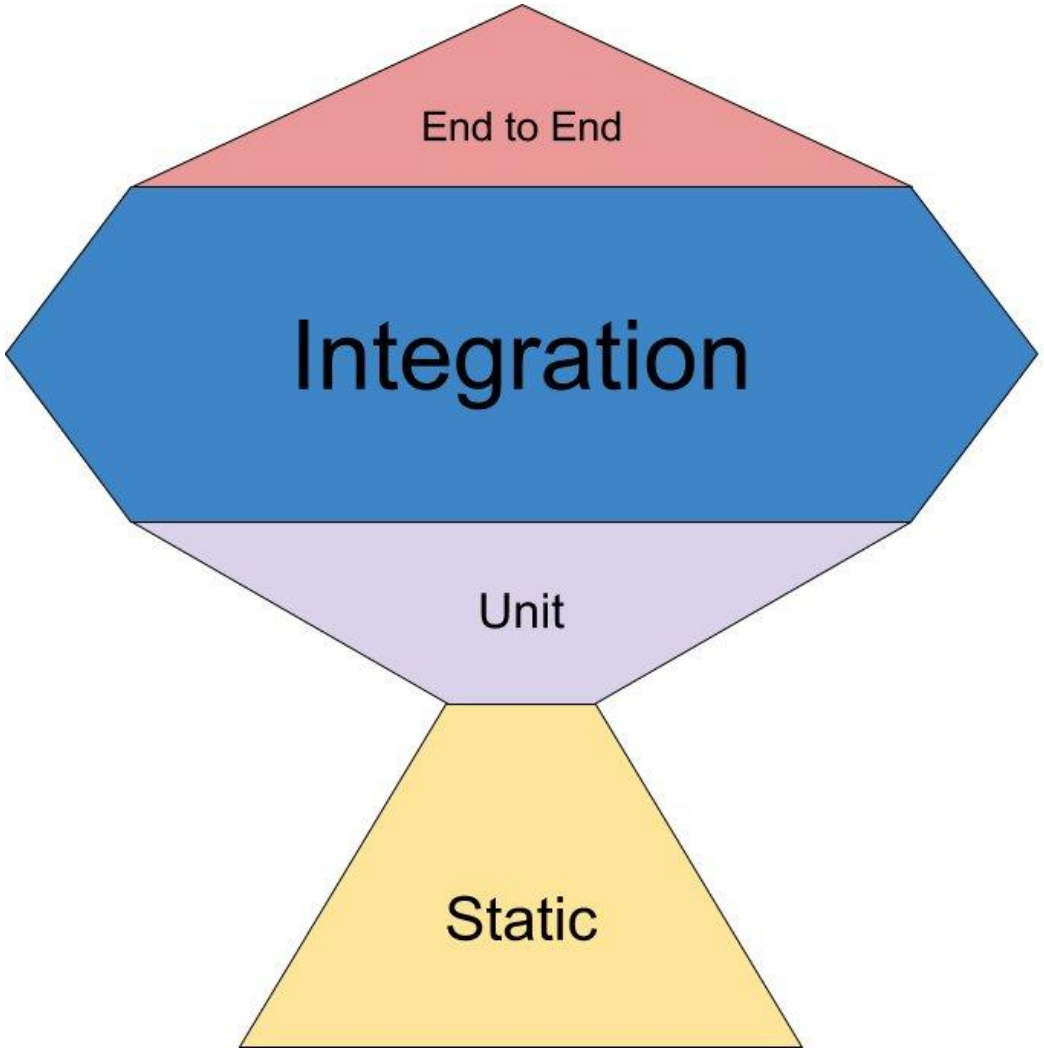
Requirements:

- services work properly
- services understand each other
- the application works with all of the services together
- the application will work if we restart some services
- application is fast enough

# Testing Pyramid



# Testing Trophy



# What strategy have we chosen?

Testing trophy:

- focus on integration tests
- check if services communicates well
- tests modules with real services

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# What testing layers we implemented to fulfill our needs?

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Implemented testing layers:

- integration tests without UI
- integration tests with Puppeteer
- unit tests
- end to end tests

# Contract tests

- check if services understand each other
- detect changes on endpoints or payloads
- does not check side effects

# Component tests

- known also as “module tests”
- test modules in isolation

# Static code tests

- rules for style checking
- configure rules
- various tools like: tslint, eslint, sonarqube
- typescript

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# Summary

- Why is QA the most important factor behind successful software projects?
  - creating high quality products
  - no shortcuts in software development
  - well defined process
- How to develop quality software and not to go bankrupt?
  - the right QA strategy (‘optimal’)
  - cost of good quality vs cost of bad quality
  - test approach (layers, quantity, available libraries)
  - test automation (ROI)
- Which types of tests will be the best for your software project?
  - Think, discuss, ask questions
  - Test Pyramid vs Testing Trophy

# Summary

- What are the practical differences between Quality Assurance and Quality Control?
  - proactive vs reactive
  - process vs product

- What is the "extended standard approach" in software testing?
  - contract tests
  - component tests
  - static code tests
  - everything to improve your coverage with low costs

# Biweekly essentials for ambitious tech leaders

Combining business and technology, this newsletter includes content hand-picked by C-level executives with a software development background.

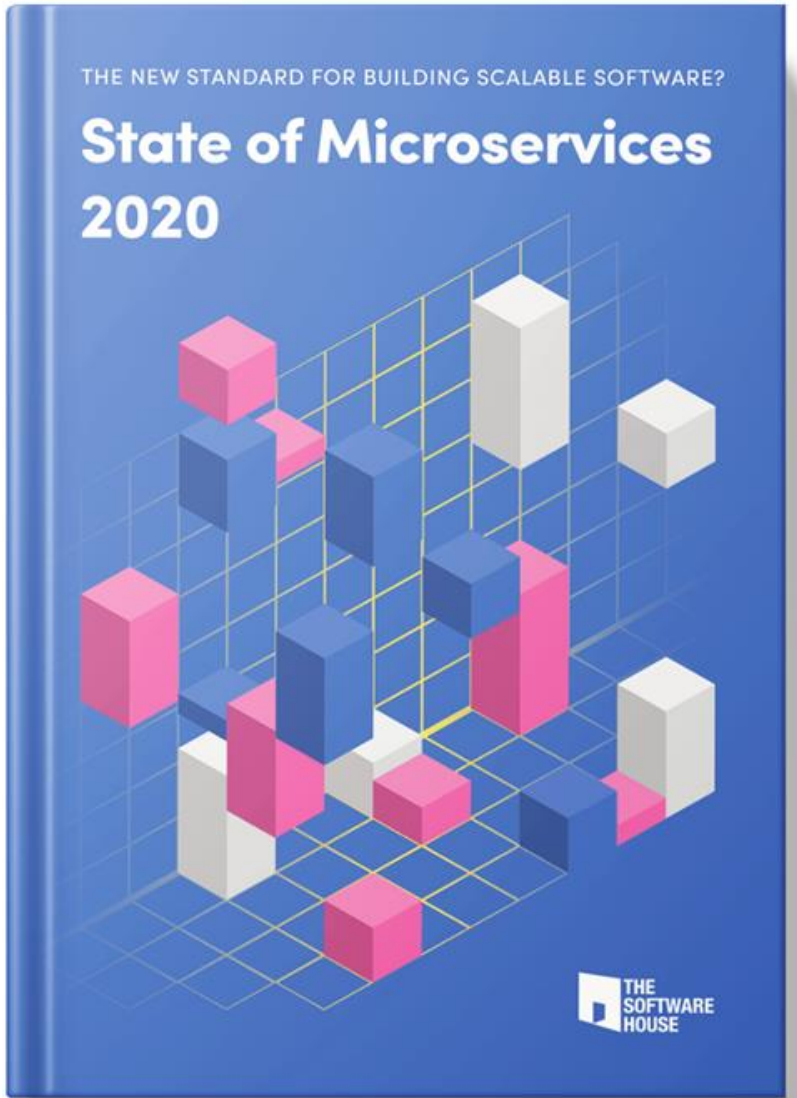


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# State of Microservices 2020

- Report based on a survey conducted among 650+ tech leaders
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# The end

Thank you